Week 5 - Part A: nine charts broken by the encoding or the geography

Syed Sami Ullah · BSAI-7B · CDB601220 Data Visualization

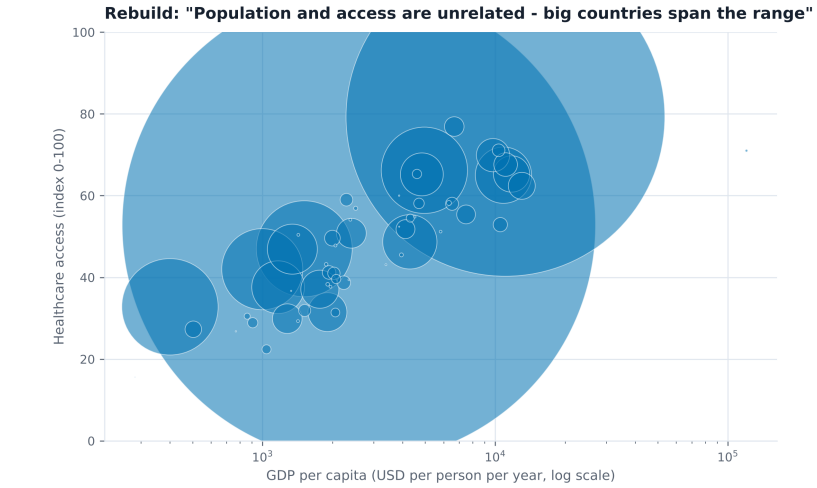
Each page: the case and the recipe, my rebuild (which hits the brief's self-check number, asserted in `partA_charts.py`), the redesign, and the diagnosis: the conclusion the chart invites, the decoy and why the data rules it out, the channel or boundary that is lying with the number that proves it. All data is synthetic course data for CDB601220 Week 5. Maps drawn with geolib in Lambert cylindrical equal-area.

The two cases fixed without changing the chart type

Case 3 and case 4. Both redesigns are still parallel coordinates plots of the same 60 lines. Case 3 changes only the scaling (log10 before min-max), case 4 changes only the axis order (greedy chain by |r|). Case 5 keeps its map too, but the fix there is to stop pretending there are five levels, so I added a strip of intervals beside it.

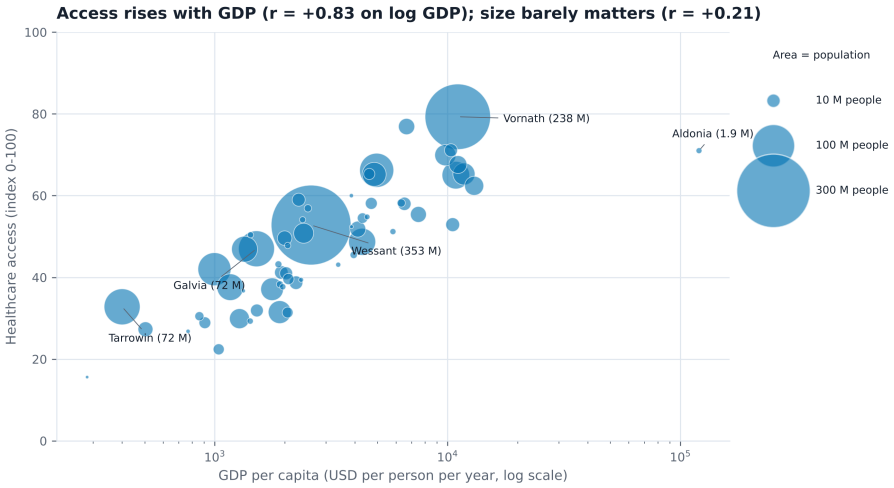
Case 1 · "Population and healthcare access are unrelated - the large countries are spread right across the range"

Recipe: `ax.scatter(df.gdp_pc, df.healthcare_access, s=df.pop_m**2)` on a log x axis.



Source: synthetic course data, CDB601220 Week 5 (health_indicators.csv). Chart: Syed Sami Ullah.
Mark size: s = pop_m**2 (the recipe). Alpha 0.55 added so covered marks show.

What this chart shows. Two huge blobs (Wessant and Vornath) take over the middle and top of the chart and a few mid-sized ones sit around them. The small countries are specks you can barely find.



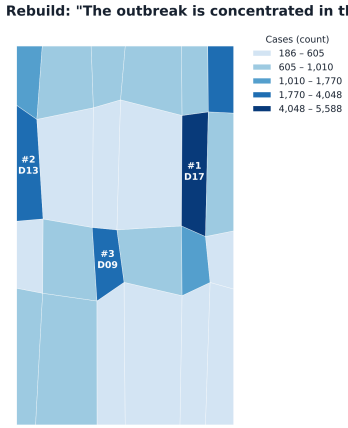
Source: synthetic course data, CDB601220 Week 5 (health_indicators.csv). Chart: Syed Sami Ullah.
Mark AREA proportional to population: s = 6 pt² x pop / 0.6 M, so the biggest is 589x the smallest in area and 24x across.

What the redesign shows. With area proportional to people and a size legend, Wessant is a big mark but not a blanket, every small country is visible, and the real story comes out: access climbs with GDP. Size is spread across access, but weakly tilted upward.

- **Conclusion it invites:** country size has nothing to do with healthcare access, because the big marks are scattered from the bottom to the top.
- **Decoy:** my first thought was plain overplotting, fix it with transparency. Ruled out: alpha only lets you see through the marks, it doesn't change their sizes. Wessant has 1.48× Vornath's people and is still drawn 2.20× the area at any alpha.
- **Mechanism:** the value drives the **radius**, not the area. `s` in `ax.scatter` is already an area (points²), so s=pop**2 makes the diameter grow with population and the area with population squared. Population spans 589×, so the largest mark is 589× the smallest across and 346,921× in area (Lie Factor 590). The two biggest marks hold 85% of all the ink while having 40% of the people, 57 of the 60 marks have their centre under a bigger mark, and the 9 countries under 2.0 M are under 2 pt across. "The large countries" ends up meaning two blobs. (The brief's self-check says "589× the area"; on my rebuild 589× is the diameter and the area ratio is 589².)
- What the data actually says: r(pop, access) = +0.21 (p = 0.11), so size and access are only weakly related. The chart happens to point near the right answer, but it can't show it, and it hides the strong pattern underneath: access rises with log GDP (r = +0.83).

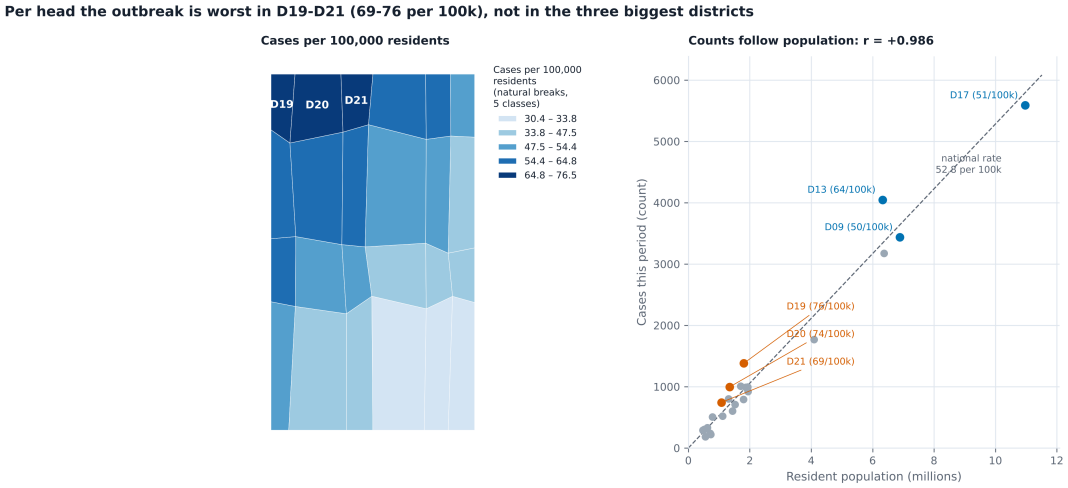
Case 2 · "The outbreak is concentrated in the three districts around the capital"

Recipe: join on code, colour districts by cases with five natural breaks and a sequential blue ramp, list the top three.



Source: synthetic course data, CDB601220 Week 5 (district_health.csv). Chart: Syed Sami Ullah.
Boundaries: districts.geojson, fictional federation (course file, Week 5, 2026). Lambert cylindrical equal-area, true scale 30.5°N.

What this chart shows. Three dark districts (D17, D13, D09) with the rest fading to pale blue. It reads like a clear hotspot, and the legend says only "cases".



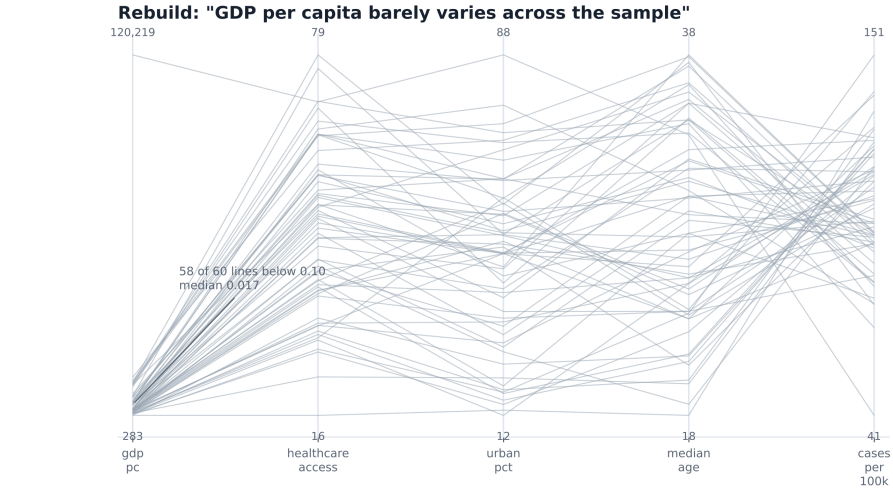
Source: synthetic course data, CDB601220 Week 5 (district_health.csv). Chart: Syed Sami Ullah.
Boundaries: districts.geojson, fictional federation (course file, Week 5, 2026). Lambert cylindrical equal-area, true scale 30.5°N. Natural breaks: 30.4, 33.8, 47.5, 54.4, 64.8, 76.5

What the redesign shows. Per resident, the dark districts move to the north-west (D19-D21). The right panel shows why the count map misled: the counts sit on the national-rate line and the big districts are simply at the far end of it.

- **Conclusion it invites:** three districts are where the outbreak is, so that's where the response should go.
- **Decoy:** my first thought was that the three are just the biggest districts, so they have more land to catch cases. Ruled out: they rank 15th, 16th and 21st of 24 by area, and r(cases, area) is −0.35. Land isn't the driver.
- **Mechanism:** the colour carries a **raw count**, so the map is a population map. r(cases, population) = +0.986, while r(rate, population) = −0.000. The three count districts hold 6.3-11.0 M people each. The top three by **rate** (D19, D20, D21, at 69-76 per 100,000) hold 1.1-1.8 M, and the two lists share **zero** districts. D17, the "worst" district, runs at 51.0 per 100,000, below the national 52.8.

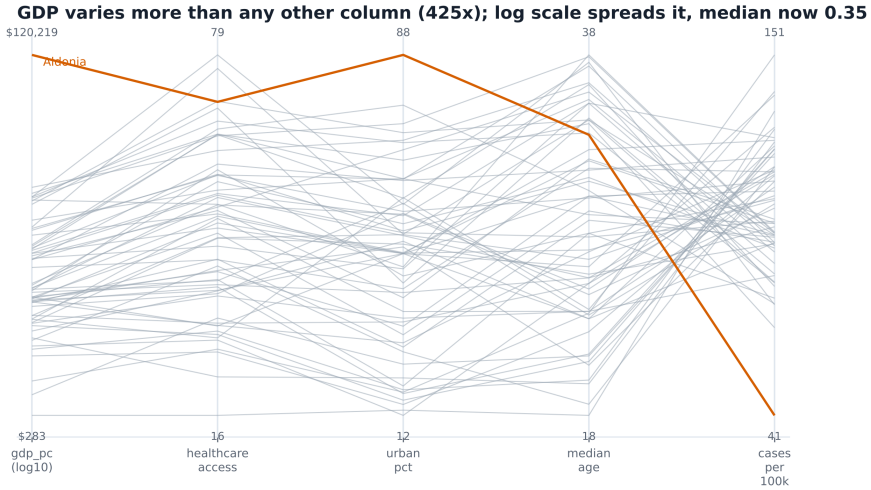
Case 3 · "GDP per capita barely varies across the sample"

Recipe: min-max scale the five numeric columns to 0-1, parallel coordinates, look at the gdp_pc axis.



Source: synthetic course data, CDB601220 Week 5 (health_indicators.csv). Chart: Syed Sami Ullah.
Axes min-max scaled to 0-1, file order. Real min and max printed at each axis end.

What this chart shows. Every line leaves the GDP axis from almost the same spot at the bottom, except one line that starts at the very top. Read literally, GDP is the one variable that doesn't vary.



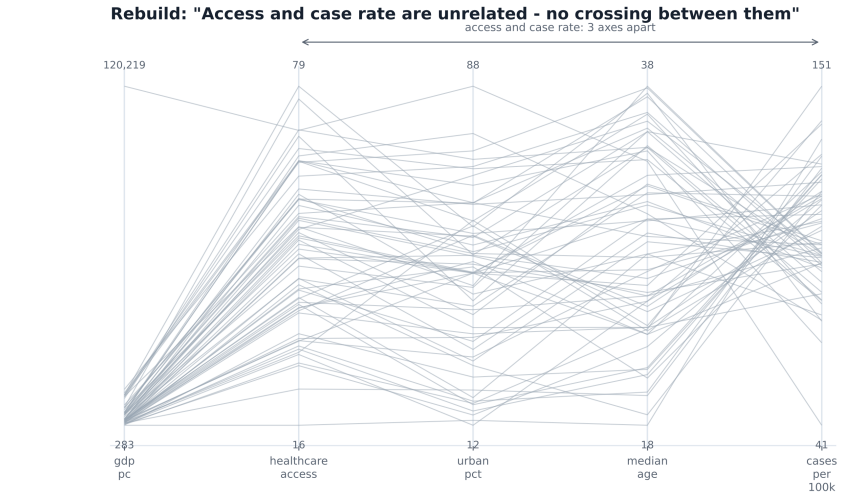
Source: synthetic course data, CDB601220 Week 5 (health_indicators.csv). Chart: Syed Sami Ullah.
gdp_pc log10-transformed, then every axis min-max scaled to 0-1. File order kept so only the scaling changed.

What the redesign shows. Same plot, same order, one decision changed: GDP goes through log10 before min-max. The GDP axis now spreads from bottom to top, Aldonia is still there (orange) but no longer sets the scale, and you can see GDP lines running roughly parallel to healthcare access.

- **Conclusion it invites:** GDP per head is about the same in every country, so it can't explain anything else in the plot.
- **Decoy:** my first thought was that Aldonia's $120,219 is a typo with an extra digit, so just delete it. Ruled out: the rest of its row fits a small rich state (88% urban, access 71, median age 33.4, 1.9 M people), and the data dictionary calls it "a real kind of country". Deleting it also doesn't fix the axis: min-max on the other 59 still puts the median at 0.16 with 16 of 59 lines under 0.10, because the next richest country is 46× the poorest.
- **Mechanism: min-max scaling** lets one value set the range. Aldonia is 9.2× the second richest, so the other 59 countries are squeezed into the bottom tenth of the axis (58 of 60 below 0.10, median 0.017). GDP actually varies the most of the five columns: 425× from poorest to richest and a coefficient of variation of 2.65, against 0.19-0.40 for the others.

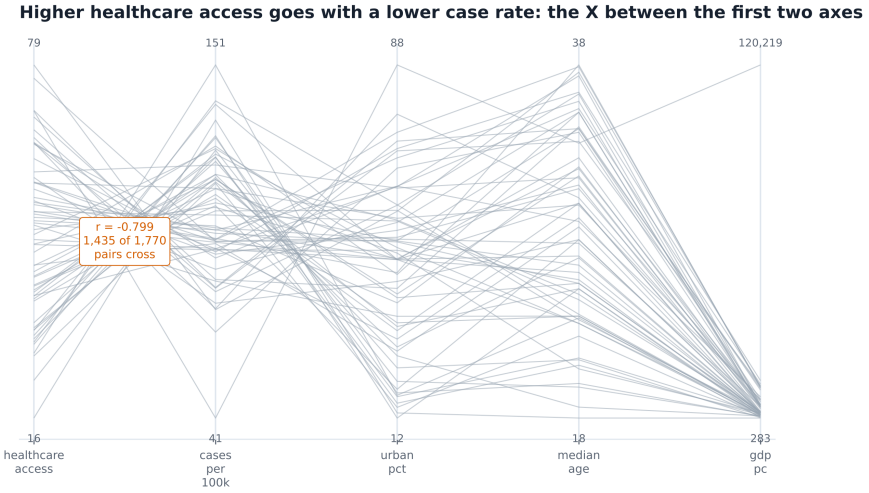
Case 4 · "Healthcare access and case rate are unrelated - there is no crossing pattern between them"

Recipe: parallel coordinates in file column order, min-max scaled, read the gaps between adjacent axes.



Source: synthetic course data, CDB601220 Week 5 (health_indicators.csv). Chart: Syed Sami Ullah.
Axes min-max scaled to 0-1, in the order the columns sit in the file.

What this chart shows. Lines mostly run in parallel between the first four axes, and the only big crossing band is between `median_age` and `cases_per_100k`. Nothing is visible between access and case rate because they never sit next to each other.



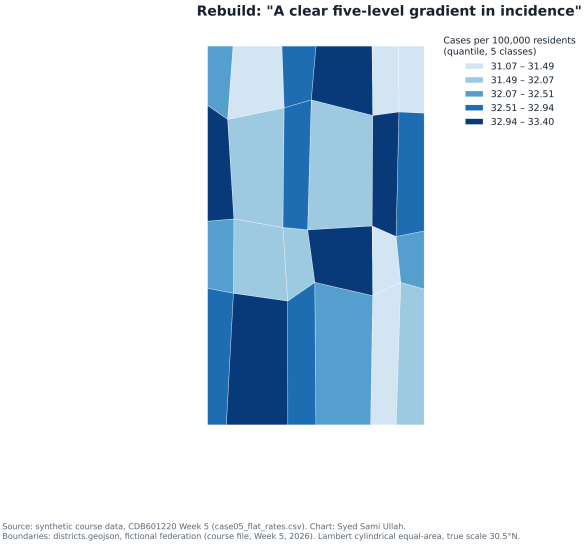
Source: synthetic course data, CDB601220 Week 5 (health_indicators.csv). Chart: Syed Sami Ullah.
Axes min-max scaled to 0-1. Order = greedy chain by |r| (slide 24), fixed before drawing: healthcare_access -> cases_per_100k -> urban_pct -> median_age -> gdp_pc.

What the redesign shows. Same lines and same scaling, only the order changed. With access and case rate side by side, the lines cross in a tight X, high access to low cases. (GDP is still flat on the far right; that's the case 3 problem, which I left alone here so the order is the only thing that changed.)

- **Conclusion it invites:** access and case rate don't move together, so improving access wouldn't change the case rate.
- **Decoy:** my first thought was that the link is there but non-linear or driven by one country, so a plot of lines wouldn't show it anyway. Ruled out: Pearson r = −0.799 and Spearman −0.810, so it's monotonic and no single outlier makes it. It's the strongest pair in the whole table.
- **Mechanism:** a parallel coordinates plot only shows the relationship between **adjacent** axes, so the **axis order** decides which pairs you can see. In file order access and case rate are three axes apart. Put next to each other they produce 1,435 crossings out of 1,770 line pairs (the X-shaped waist). The file order spends its four adjacent slots on pairs with |r| = 0.36, 0.75, 0.64 and 0.59 and leaves the strongest pair in the table (0.80) out.

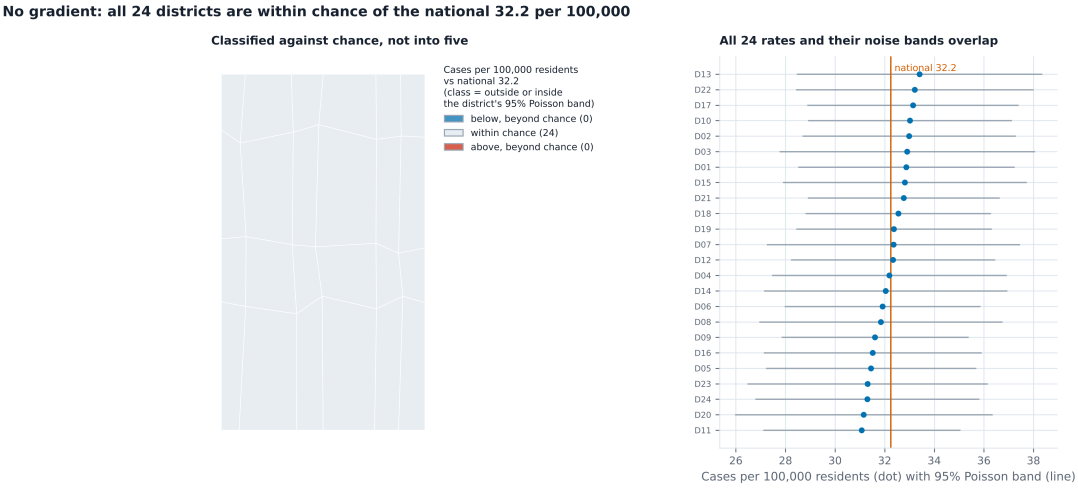
Case 5 · "There is a clear five-level gradient in incidence across the federation"

Recipe: `cases / population * 1e5`, five quantile classes, five-step sequential ramp.



Source: synthetic course data, CDB601220 Week 5 (case05_flat_rates.csv). Chart: Syed Sami Ullah.
Boundaries: districts.geojson, fictional federation (course file, Week 5, 2026). Lambert cylindrical equal-area, true scale 30.5°N.

What this chart shows. A confident five-shade map with dark and pale districts scattered across the federation. The legend gives it away: the whole scale runs from 31.07 to 33.40.



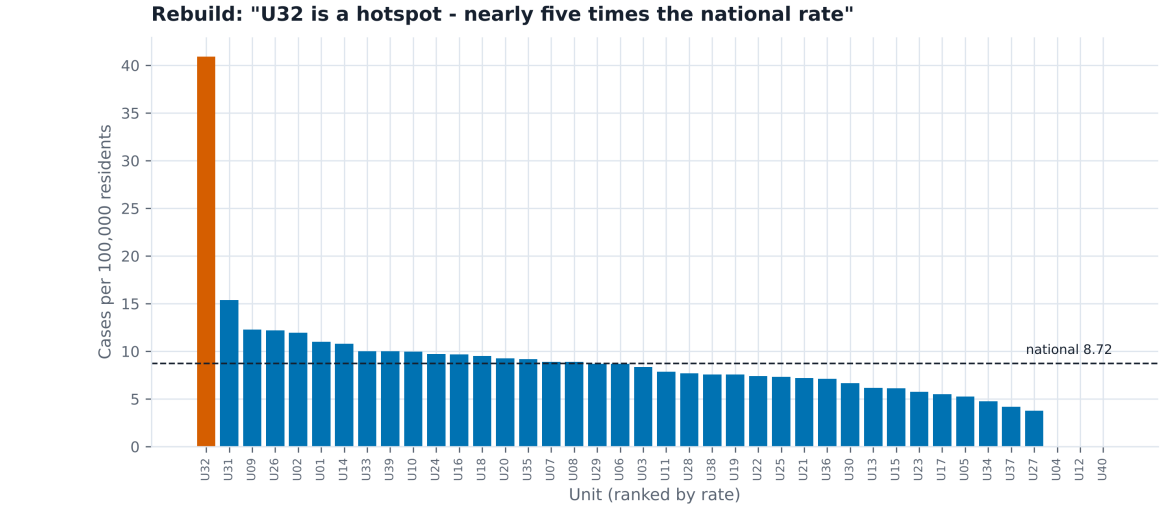
Source: synthetic course data, CDB601220 Week 5 (case05_flat_rates.csv). Chart: Syed Sami Ullah.
Boundaries: districts.geojson, fictional federation (course file, Week 5, 2026). Lambert cylindrical equal-area, true scale 30.5°N. Diverging ramp centred on the national rate (a real midpoint).

What the redesign shows. Classified against chance instead of into five equal bins, every district lands in the same class and the map is one flat colour. That's the honest picture. The strip on the right shows why: every district's noise band crosses the national line.

- **Conclusion it invites:** incidence varies in five clear steps across the federation, so some districts need more attention than others.
- **Decoy:** my first thought was that quantile is just the wrong scheme and Jenks would fix it. Ruled out: natural breaks and equal interval paint the same five-level map, with GVF 0.968 and 0.961 against quantile's 0.959. Any five-class scheme will fill five classes. The scheme isn't the problem.
- **Mechanism:** the **classification** turns a 2.3-point spread into five colours. The rates run 31.07-33.40 (7.2% of the mean), so each quantile class is about 0.47 per 100,000 wide, around 1.4%. Poisson noise alone would move a district of median size by ±4.4 per 100,000 at 95%, almost twice the whole observed spread. 0 of 24 districts fall outside their band, and a chi-square test against one shared rate gives 2.30 on 23 df (p = 1.00). The districts are more alike than chance would make them. There is no gradient to classify.

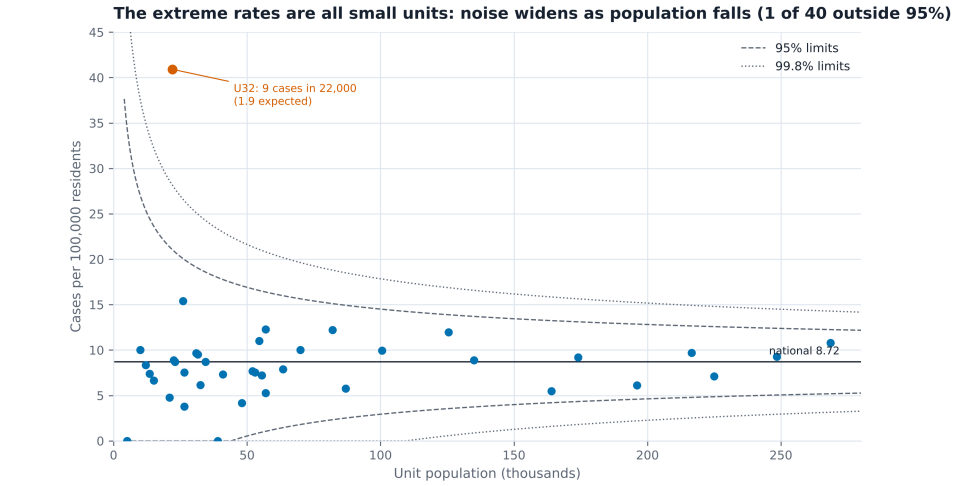
Case 6 · "Unit U32 is a hotspot - it is running at nearly five times the national rate"

Recipe: `cases / population * 1e5` per unit, rank them, bar-chart, compare the worst unit to the pooled national rate.



Source: synthetic course data, CDB601220 Week 5 (smallarea_cases.csv). Chart: Syed Sami Ullah.

What this chart shows. One orange bar towering over the rest at 40.9, almost five times the dashed national line. Nothing on the chart says how many people are behind each bar.



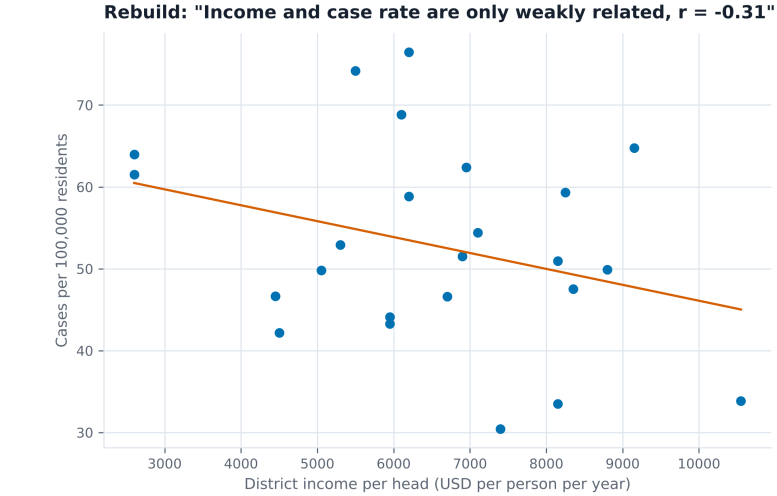
Source: synthetic course data, CDB601220 Week 5 (smallarea_cases.csv). Chart: Syed Sami Ullah.
Funnel: Poisson limits around the pooled rate, mean +/- z*sqrt(expected) (geolib.funnel_limits).

What the redesign shows. Plotted against population, the rates form a funnel: wild at the small end, tight at the big end. U32 is the one unit poking out, at the narrow-denominator end, and the zero-rate units on the floor are just as small.

- **Conclusion it invites:** something is going on in U32, which is running at 4.7× the national rate and needs an investigation team.
- **Decoy:** my first thought was that the comparison is unfair: the pooled rate is dominated by the big units, so U32 should be compared to the average unit. Ruled out: the plain mean of the 40 unit rates is 8.58, close to the pooled 8.72, so U32 is still 4.8× that. The baseline isn't the issue.
- **Mechanism:** a **small denominator**. U32's rate rests on 9 cases in 22,000 people. At the national rate it would expect 1.9 cases, and every single case moves its rate by 4.5 per 100,000, half the national rate. Its exact 95% interval runs from 18.7 to 77.7. The ranking puts small units at **both** ends: the top two (U32, U31) are under 30,000 people and the bottom three are zeros from units of 5,000-39,000, while the 10 units over 100,000 all sit between 5.5 and 12.0. The file has one true rate everywhere, so the spread is noise that grows as population shrinks. In the funnel, 1 of 40 units is outside the 95% limits, fewer than the 2 you'd expect by chance from 40 units. U32 is unusual ($p \approx 0.007$ even after allowing for 40 looks), but it is still nine cases. That's a reason to check the records, not to publish a hotspot.

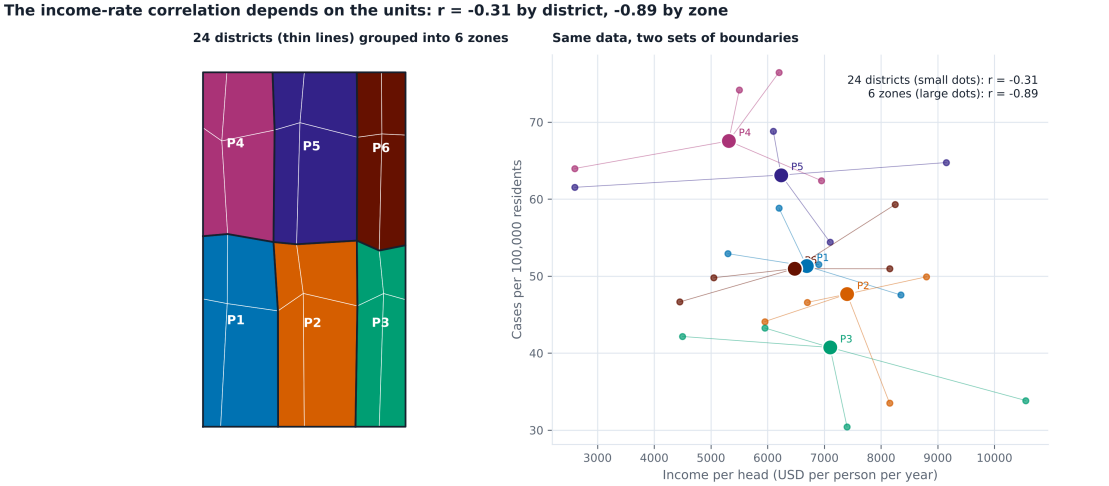
Case 7 · "District income and case rate are only weakly related - r is about −0.31"

Recipe: correlate `income_pc_usd` with `cases / population * 1e5` across the 24 districts.



Source: synthetic course data, CDB601220 Week 5 (district_health.csv). Chart: Syed Sami Ullah.

What this chart shows. A loose cloud of 24 districts with a gently falling line through it. Weak, noisy, not much to say.



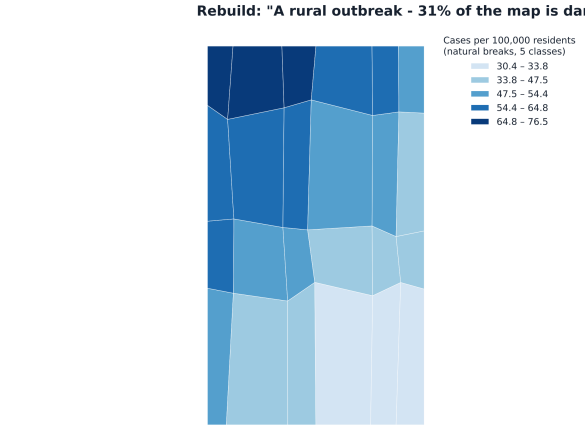
Source: synthetic course data, CDB601220 Week 5 (district_health.csv). Chart: Syed Sami Ullah.
Boundaries: districts.geojson, fictional federation (course file, Week 5, 2026). Lambert cylindrical equal-area, true scale 30.5°N. Zones from districts_coarse.geojson. Zone income = mean of its 4 districts; zone rate = pooled cases / pooled population.

What the redesign shows. Both scales on one chart, with lines tying each district to its zone. Grouped into zones, the noise inside each zone averages out and the slope looks steep. The honest statement is both numbers together, each with its unit.

- **Conclusion it invites:** income barely matters for the case rate, r ≈ −0.31, so there's no point targeting poorer areas.
- **Decoy:** my first thought was that one odd district is dragging r around. Ruled out: dropping each district in turn keeps r between −0.39 and −0.19. No single district makes or breaks it.
- **Mechanism:** the **areal unit** sets the coefficient (MAUP, scale effect). The same cases, people and incomes grouped into the six coarse zones give r = −0.889. Inside the zones r runs from −0.69 to +0.77. Even the way income is aggregated matters: population-weighted zone income gives −0.746. So "−0.31" isn't a fact about income and illness, it's a fact about these 24 boundaries. And neither number says anything about individual people (the ecological fallacy).

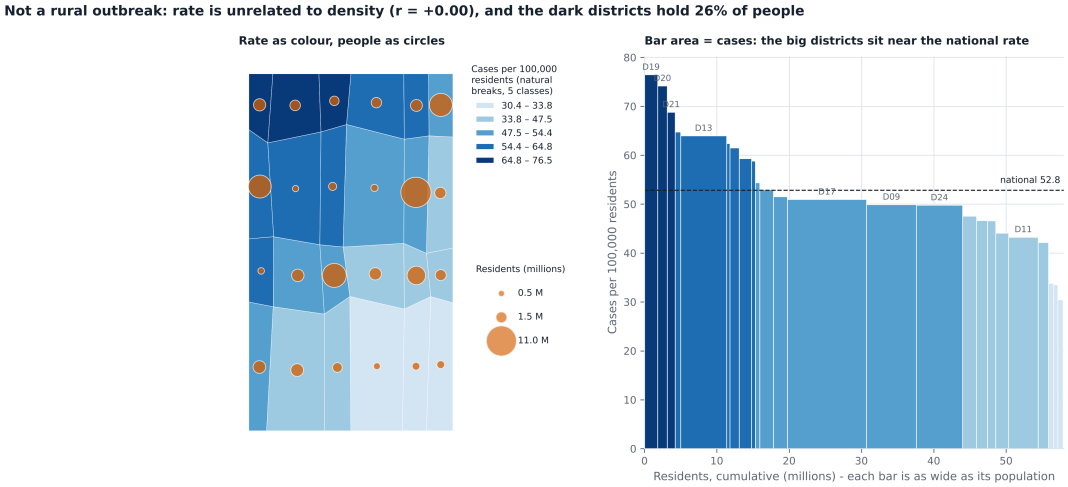
Case 8 · "This is a rural outbreak - look how much of the map is dark"

Recipe: choropleth of the rate, five natural breaks, then describe the map by how much of it is in the top two classes.



Source: synthetic course data, CDB601220 Week 5 (district_health.csv). Chart: Syed Sami Ullah.
Boundaries: districts.geojson, fictional federation (course file, Week 5, 2026). Lambert cylindrical equal-area, true scale 30.5°N.

What this chart shows. The whole northern band and some of the middle are dark, and big pale districts fill the south. The eye reads "lots of dark land", and the dark land is the sparse part of the country.



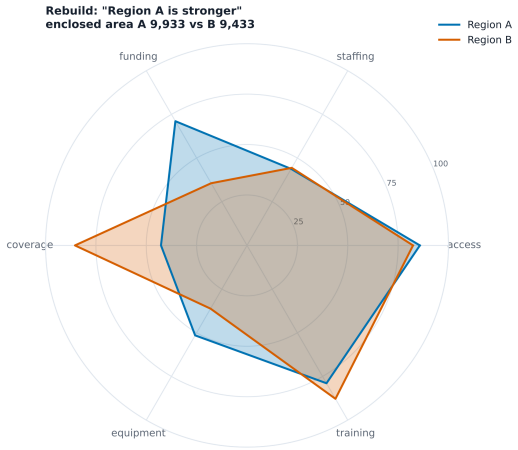
Source: synthetic course data, CDB601220 Week 5 (district_health.csv). Chart: Syed Sami Ullah.
Boundaries: districts.geojson, fictional federation (course file, Week 5, 2026). Lambert cylindrical equal-area, true scale 30.5°N. Circle AREA proportional to residents.

What the redesign shows. With the people put back (circles on the map, bar widths on the right), the epidemic's weight is in the five big districts near the national rate. The high-rate districts are the thin bars on the left: real, worth a look, but a small share of the people.

- **Conclusion it invites:** the outbreak is a rural problem, because most of the dark ink is on big, empty northern districts.
- **Decoy:** my first thought was the projection: a Mercator-style map swells the north, which is exactly where the dark districts are. Ruled out: this map is equal-area, and even Mercator would only make the north 1.49/1.24 ≈ 1.2× too big relative to the south, nowhere near enough to manufacture the picture.
- **Mechanism: area read as people**. A choropleth gives each district ink in proportion to its land, but the rate is about residents. The five most populous districts are 12.6% of the land and 59.9% of the people (58.9% of the cases), so most of the epidemic is drawn as small patches. And the "rural" reading fails on the numbers: r(rate, density) = +0.00, and the 12 least dense districts run at 52.0 per 100,000 against 53.0 for the 12 most dense. The dark classes cover 31% of the land and hold 26% of the people, and they are dark because of their rate, not because they're rural.

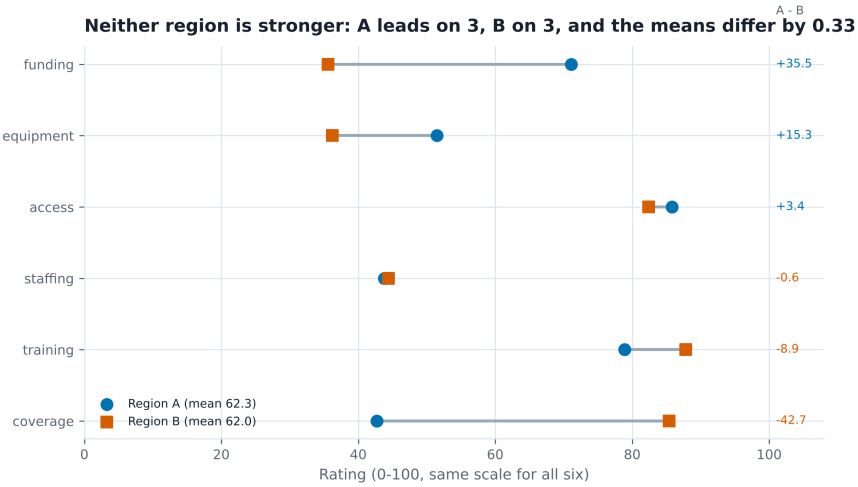
Case 9 · "Region A is the stronger of the two - its profile encloses visibly more area"

Recipe: radar chart, six axes in file order, both profiles on one pair of axes, filled. Compare the enclosed areas.



Source: synthetic course data, CDB601220 Week 5 (region_profiles.csv). Chart: Syed Sami Ullah.
Radar: six axes in file order, same 0-100 scale, filled at alpha 0.25.

What this chart shows. Blue (A) wraps around a bit more than orange (B), and with the fill it looks like A is the bigger, stronger shape. Each region has three spikes.



Source: synthetic course data, CDB601220 Week 5 (region_profiles.csv). Chart: Syed Sami Ullah.
Indicators sorted by the A - B gap. Squares and circles so the regions differ by shape as well as colour.

What the redesign shows. Laid out as a dot plot, position carries the values and nothing depends on axis order. A is far ahead on funding and equipment, B is far ahead on coverage and training, and overall they're level.

- **Conclusion it invites:** Region A performs better overall, because its shape is bigger.
- **Decoy:** my first thought was that A really is a bit better on average. Ruled out: the means are 62.3 and 62.0, a gap of 0.33 on a 0-100 scale, and A leads on 3 indicators while B leads on the other 3. Small gaps on some and big gaps both ways on the others (funding +35.5 for A, coverage +42.7 for B).
- **Mechanism:** the **enclosed area** of a radar polygon is $\frac{1}{2}\cdot\sin(60°)\cdot\Sigma\, v_i \cdot v_{i+1}$, a sum of products of neighbouring values. So it depends on which indicators sit next to each other, which is just the file order. A's two strengths (access, training) happen to sit beside each other through the wrap-around. Across the 60 distinct axis orders, A encloses more in 39 and B in 21, and the gap swings from −1,131 to +1,659. In the order access, staffing, funding, equipment, training, coverage, B wins 10,497 to 9,366.